

# John Paul R. Martinez

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## EDUCATION

**Manuel S. Enverga University Foundation | Lucena City, Philippines**

**May 2027**

**B.S. in Computer Science Major in Software Engineering**

- **Relevant Courses:** Programming Languages, Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering
- **Honors:** Dean's Lister

## TECHNICAL & RESEARCH PROJECTS

**KILATIS | Digital Image Forensics Framework**

**January 2026 - Present**

*Thesis Project*

- Built a dual-branch deep learning forensic framework using PyTorch, TruFor (Transformers), and FFT/DWT analysis to detect AI-Generated / deepfakes and generate pixel-level tampering heatmaps for traditional splices.
- Engineered a cross-platform forensic suite using FastAPI, Next.js 16, and Tauri v2, integrating SHA-256 hashing, EXIF parsing, and automated PDF report generation.

**Ephnyr AI | Full-Stack Multi-Tenant RAG Platform**

**August 2026**

*AI & System Architecture*

- Architected a decoupled client-server RAG platform utilizing LangChain chunking, local CPU FastEmbed (384-dim dense vectors), and PostgreSQL pgvector HNSW indexing for high-precision semantic document similarity search.
- Integrated Groq LPU inference (openai/gpt-oss-20b) to deliver AI chat completions with document citations, enforced by a zero-trace ephemeral session model and room-level storage quotas.

**FocusFolio | AI Study Workspace & Document Quiz Generator**

**July 2026**

*EdTech & AI Platform*

- Engineered computerized document processing pipelines using Pinecone and LangChain to dynamically generate interactive multiple-choice quizzes and front-to-back flashcards from uploaded course files.
- Built a context-restricted RAG study workspace featuring a side-by-side native PDF reader and an AI tutor strictly constrained to source materials to eliminate hallucinations.

**CheckMatePH | Political Fact-Checking & Civic Engagement Platform**

**May 2026**

*SIKAPTala 2026 Hackathon*

- Developed an systematized AI-driven fact-checking pipeline evaluating submitted claims against verified public records to combat online misinformation for Filipino voters.
- Implemented real-time interactive debate rooms and live audience voting using Supabase Realtime, alongside politician integrity scorecards and role-based permissions.

## LEADERSHIP EXPERIENCE

**MSEUF Association of Multimedia Artists and Technologists**

4th Year Representative | BSCS

**July 2026 - Present**

3rd Year Representative | BSCS

**July 2025 - May 2026**

- Acted as the primary representative between CS students and executive officers, communicating academic concerns and coordinating department-wide technical workshops/activities.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript

**Frameworks & Libraries:** React, Next.js, FastAPI, Node.js, Tailwind CSS, LangChain

**Databases & Backend:** Supabase, Firebase, PostgreSQL, RESTful APIs

**Developer Tools:** Git, GitHub, Docker, Postman, CI/CD basics